This is an experimental SPS-inspired candidate, not an algorithm or convergence theorem established by the MomSPS paper. It updates one matrix parameter with no weight decay or Nesterov term. Define $\langle A, B\rangle_F = \sum_{i,j} A_{ij}B_{ij}$.

Algorithm 1 SPS-inspired Muon candidate with a direction safeguard

Require: W_0, $0 \le \beta < 1$, $c \ge 1$, $\eta_{\max} > 0$, T
Require: Valid batch loss lower bounds ℓ_t
Ensure: W_T

$B_{-1} \leftarrow 0$
for $t = 0, \dots, T-1$ **do**
  Sample a mini-batch S_t
  $L_t \leftarrow f_{S_t}(W_t)$
  $G_t \leftarrow \nabla_W f_{S_t}(W_t)$
  $\Delta_t \leftarrow L_t - \ell_t$
  $B_t \leftarrow \beta B_{t-1} + G_t$
  if $\|G_t\|_F = 0$ **then**
    $W_{t+1} \leftarrow W_t$
    continue
  end if
  if $\|B_t\|_F = 0$ **then**
    $D_t \leftarrow G_t$
  else
    $D_t \leftarrow \mathrm{NewtonSchulz5}(B_t)$
  end if
  $a_t \leftarrow \langle G_t, D_t\rangle_F$
  if $a_t \le 0$ **then**
    $D_t \leftarrow G_t$ ▷ Experimental gradient fallback
    $a_t \leftarrow \|G_t\|_F^2$
  end if
  $\eta_t \leftarrow \min\{\Delta_t/(ca_t), \eta_{\max}\}$
  $W_{t+1} \leftarrow W_t - \eta_t D_t$
end for